AP CSP 4.2 Fault Tolerance

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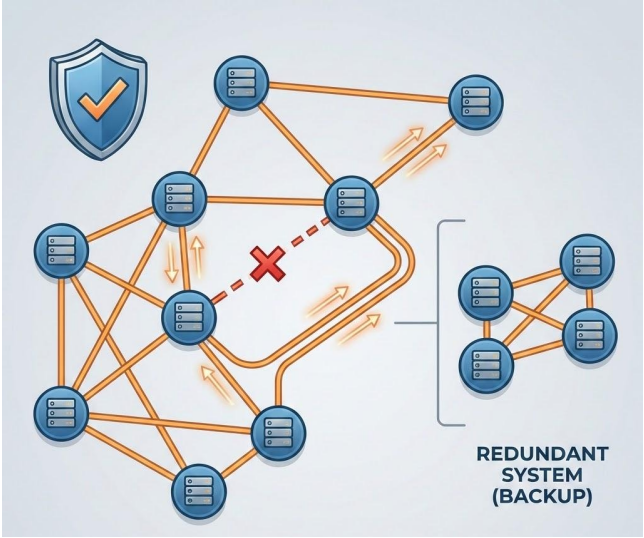
What is Fault Tolerance? - General Overview

Fault Tolerance is a system's ability to continue operating uninterrupted despite the failure of one or more of its components.

Learning Objective:

In this part of the Big Idea you should be able to explain how a system/network is fault tolerant

For a system or network to be fault tolerant ,*aka be able to function even if some components fail*, that system -or network- has to be *redundant*.



- Although in almost all cases redundancy is necessary for fault tolerance it can take up a lot of additional resources thus turning into a setback.

+ Redundancy can be achieved by having a large number of pathways between connections in networks.
+ You can add more and more devices to your network as it makes the network stronger!!

Real World Example #1

The internet!

Real World Example #2

Cloud Systems

Real World Example #3

Space Systems

Real World Example #4

Real World Example #5

Software Systems

What do they all have in common?

Redundancy

Failover mechanisms

Which of the following is NOT a benefit of a fault-tolerant network?

A. Data has more than one path to travel from one device to another.
B. If part of the network fails, the network can still function by using other paths.
C. Data will only take one route from one device to another, no matter the number of routes available.
D. More devices creates more connections and makes the network stronger.

Example Question #1

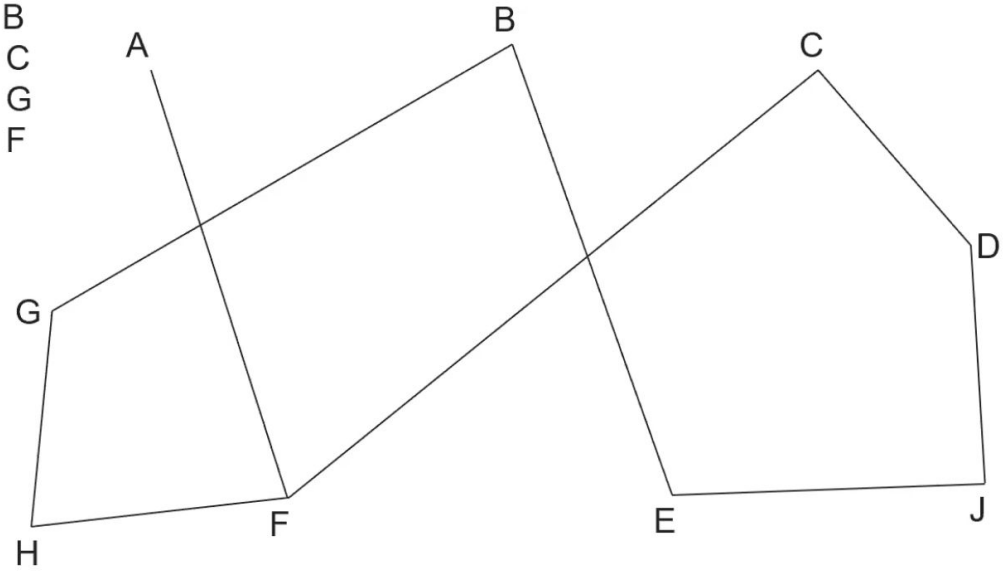
Solution

Which of the following is NOT a benefit of a fault-tolerant network?

A. Data has more than one path to travel from one device to another.
B. If part of the network fails, the network can still function by using other paths.
C. Data will only take one route from one device to another, no matter the number of routes available.
D. More devices creates more connections and makes the network stronger.

What would make this network fault-tolerant?

A. A connection from A to B
B. A connection from B to C
C. A connection from E to G
D. A connection from E to F



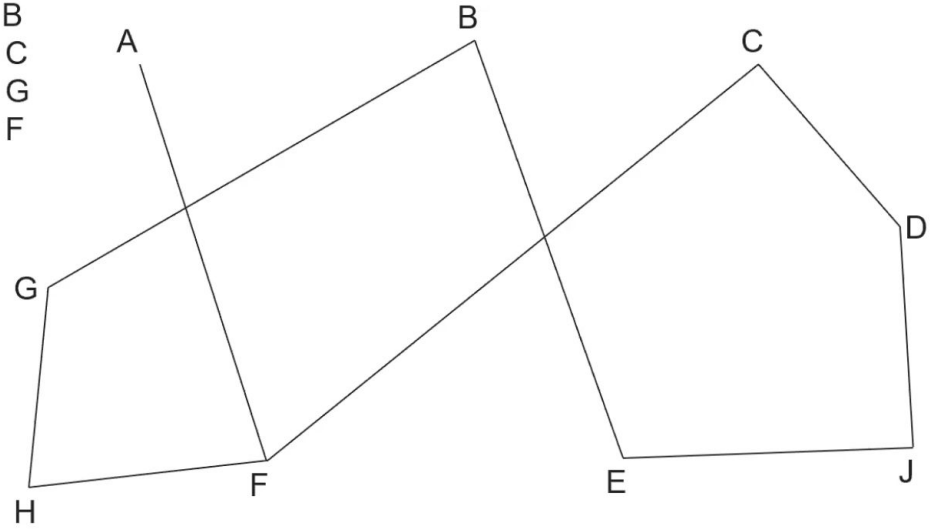
Example Question #2:

Solution

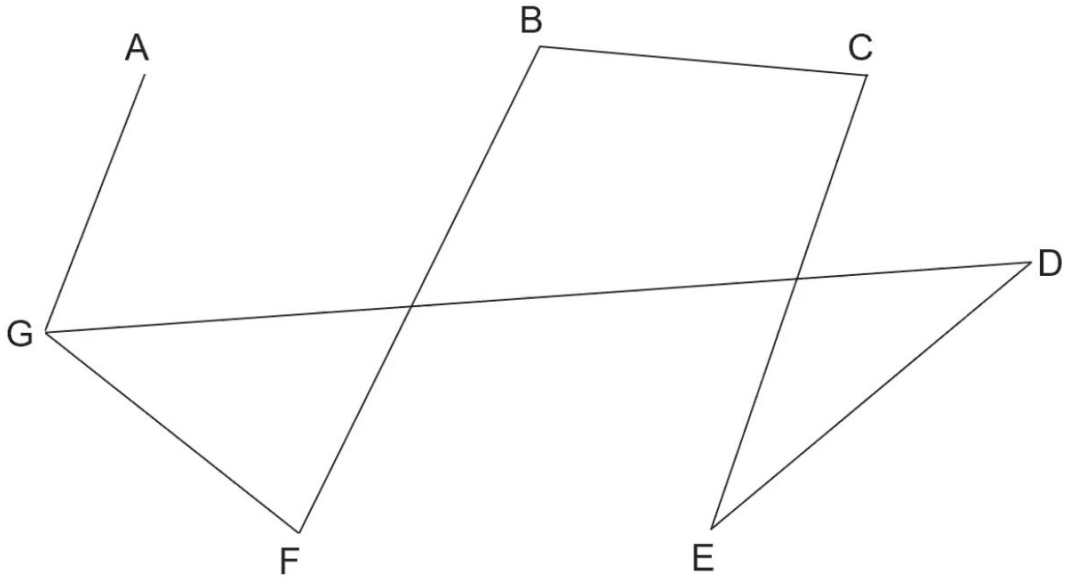
Tip: in such questions count the number of connections each points have and add a new connection to the point with the least amount of connections!

What would make this network fault-tolerant?

A. A connection from A to B
B. A connection from B to C
C. A connection from E to G
D. A connection from E to F



Is this network fault-tolerant?

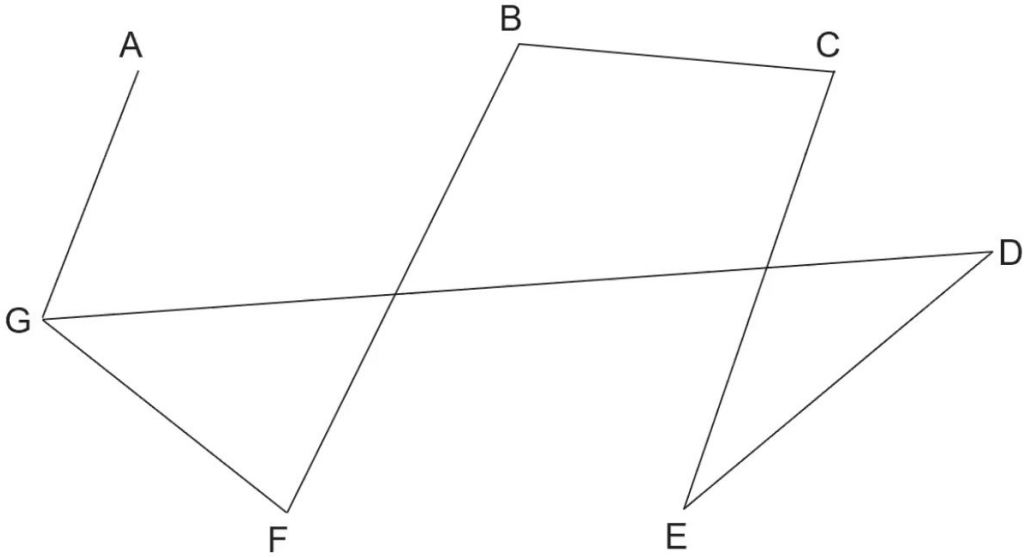


Example Question #3:

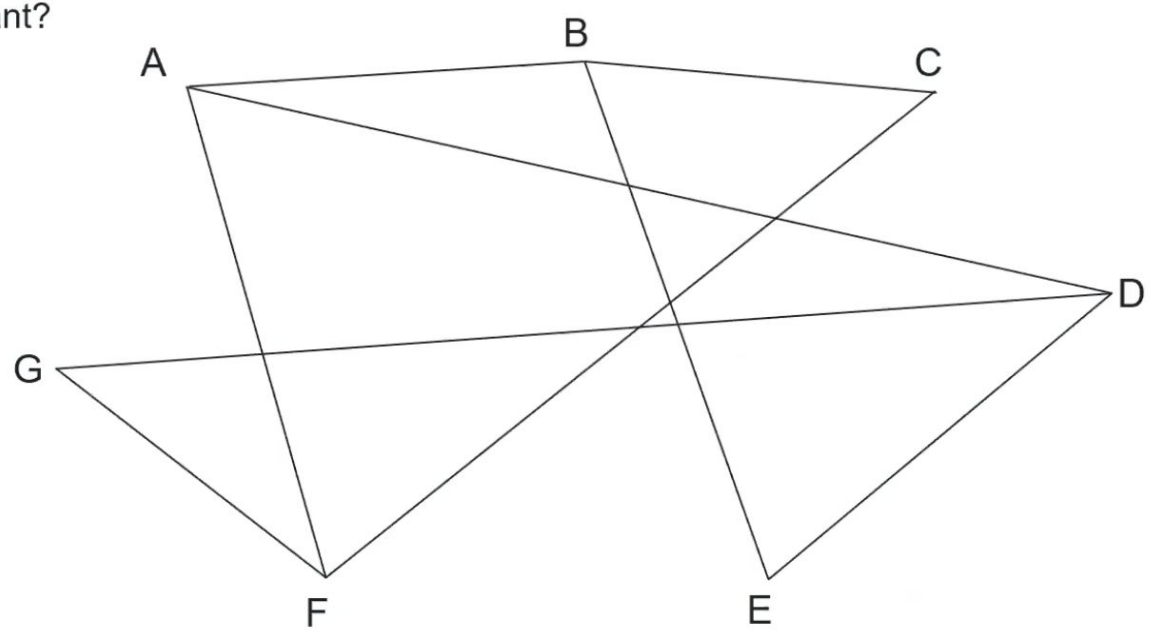
Solution

Is this network fault-tolerant?

No! It is not because if the connection between points A and G shut down there will be no connection from A to any other points causing failures in the system. If there was any other connections from point A to any other points than the network would be more fault tolerant.



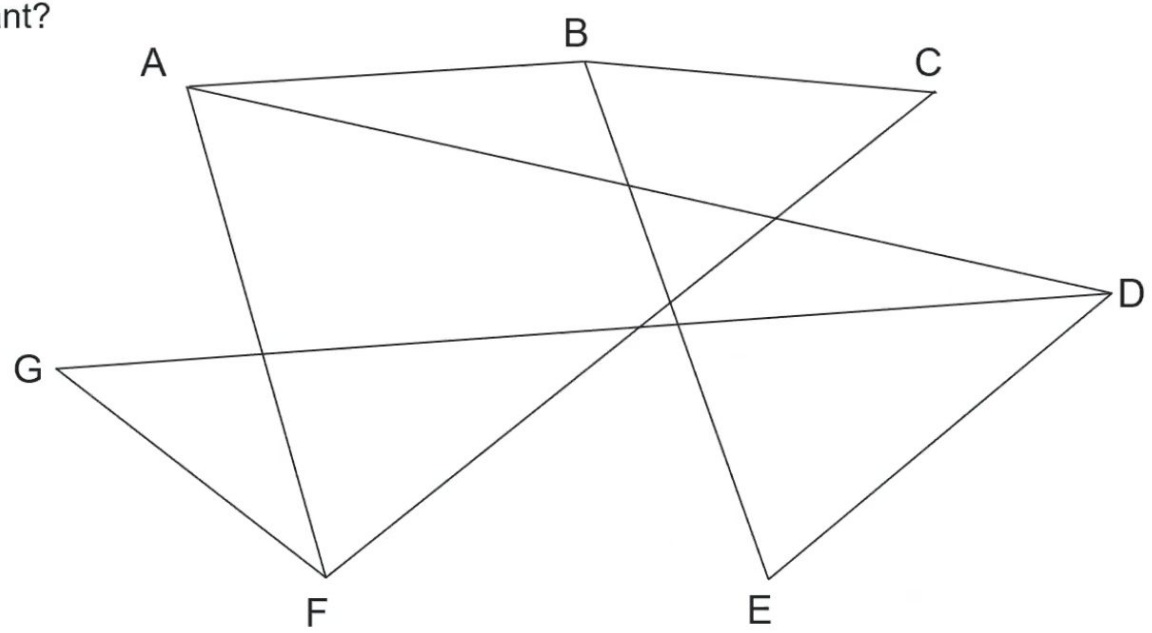
Is this network fault-tolerant?



Example Question #4:

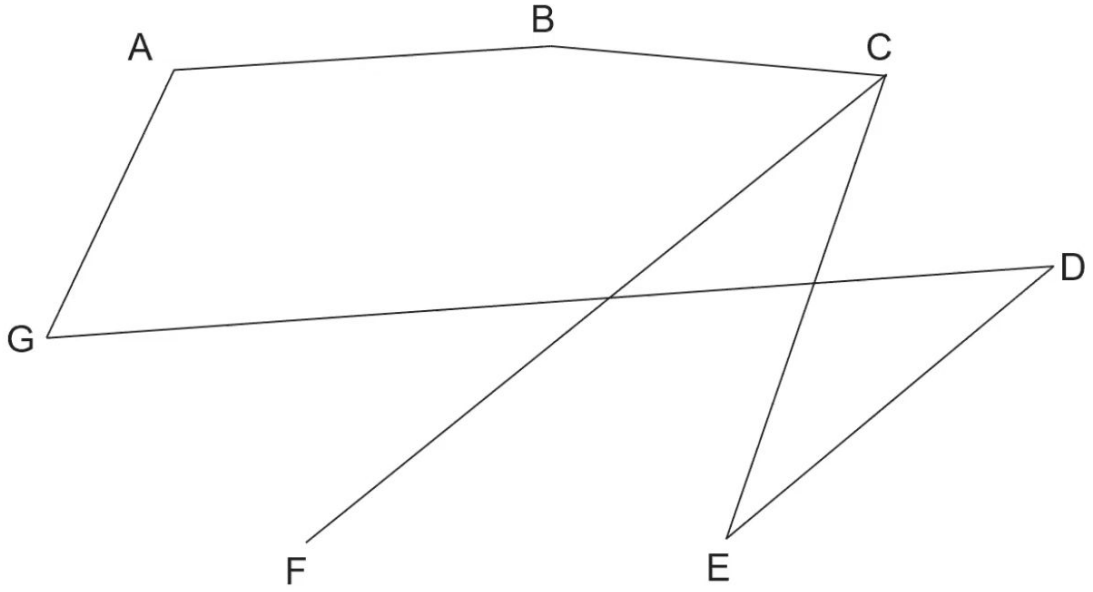
Solution

Is this network fault-tolerant?



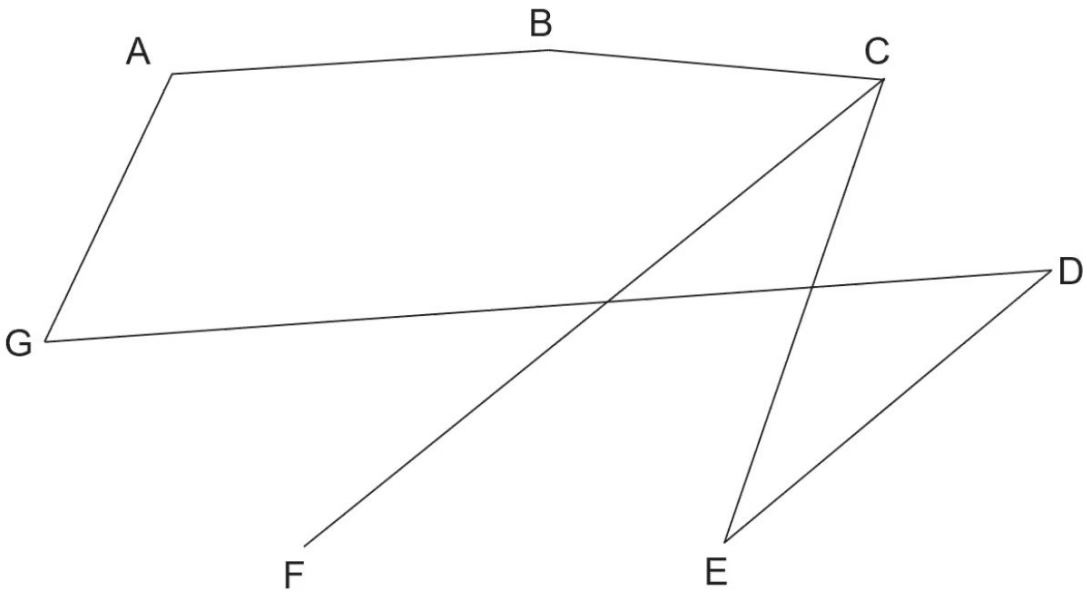
Yes it is!!
All points have at least two connections so the network is able to keep on working even if one connection shuts down.

Is this network fault-tolerant?



Example Question #5:

Solution

Is this network fault-tolerant?

No! It is not because if the connection between points F and C shut down there will be no connection from F to any other points causing failures in the system. If there was any other connections from point F to any other points than the network would be more fault tolerant.